EDUCATOR GUIDE #1

AI Search & Medical Hallucinations

Understanding Next-Token Prediction & Teachable Verification Protocols

Teacher Background: Why AI "Hallucinates" Health Facts

Large Language Models (LLMs) used in tools like Google AI Overviews and ChatGPT are not database engines or truth engines—they are **probabilistic text predictors**. They generate text token by token based on statistical patterns found in their training data. When queried about medical symptoms or treatments, an LLM selects words that sound structurally authoritative, regardless of factual correctness. This frequently leads to **confident medical falsehoods** and plausible but entirely fabricated citations (PMIDs, DOI numbers, and imaginary journal volume references).

How AI Generates Fake Citations

Because scientific literature references follow strict formatting patterns (e.g., *Author, A. (Year). Title. Journal of Medicine, Vol(Issue)*), language models master the **syntax** of academic citations long before attaining factual accuracy. When pressed for evidence, the model generates strings that look like valid peer-reviewed studies.

3-Step Student Cross-Reference Checklist

Use this structured protocol in class to teach students how to verify any AI-generated medical summary.

1. **Isolate Specific Claims & Identifier Tokens**
   Instruct students to highlight named entities: cited authors, study titles, sample sizes, publication years, or specific PMID / DOI numbers in the AI search summary.

2. **Direct Database Lookup (PubMed / WHO / Cochrane)**
   Students copy exact titles, PMIDs, or primary claim terms directly into PubMed.gov or the World Health Organization site—bypassing commercial search engines entirely.

3. **Evaluate Abstract vs. AI Summary**
   If the study exists, compare the paper's actual conclusion in the official abstract against the AI's claims. Watch for hallucinated sample sizes, inverted statistical results, or generalized findings.

15-Minute Class Debate Plan

Debate Topic: "Should AI Search Engines Be Used for Personal Medical Advice?"

Split the class into **Affirmative** (AI increases accessibility to health info) and **Negative** (AI risks patient harm via plausible falsehoods).

- **Opening Prompt:** "If an AI overview gives a clear, professional-sounding summary of medication side effects with citations, why is it dangerous to accept it without checking?"
- **Affirmative Angle:** "How might AI search help patients who live far from medical clinics or lack health insurance?"
- **Negative Angle:** "What responsibility do AI tech companies have when a user relies on a hallucinated dosage or misdiagnosed symptom?"
- **Synthesis Question:** "What guardrails or mandatory disclaimers should search platforms implement on health queries?"

Comparing Information Sources

Feature	AI Search Overviews	Primary Databases (e.g. PubMed)
Generation Method	Statistical next-word prediction	Peer-reviewed clinical research papers
Citation Reliability	**High risk of hallucination**	**Indexed & verified journals**
Accessibility	Instant plain language synthesis	Requires academic search skills
Primary Purpose	Conversational summary	Archival scientific record

Classroom Activity Wrap-Up & Key Takeaways

Conclude the session by reminding students that while AI can be an efficient starting point for brainstorming terms, it must **never** act as an authoritative medical reference. Teaching students to double-check AI output against indexed clinical databases builds lifelong information literacy skills for the digital age.

- National Center for Biotechnology Information (NCBI). PubMed Database. Accessible at pubmed.ncbi.nlm.nih.gov
- World Health Organization (WHO). Health Topics & Clinical Guidelines. Accessible at who.int